

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 8

Subject Code & Name: ET - 1 - BP807ET - COMPUTER AIDED DRUG DESIGN

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- | | |
|--|---|
| A) Define the terms (i)Contour Plots (ii) ComSIA | 2 |
| B) What is drug design? | 2 |
| C) Define the terms Bioinformatics and chemoinformatics. | 2 |
| D) Define the terms (1) Visualization (ii) Molecular Mechanics | 2 |
| E) Explain Hansch analysis. | 2 |
| F) Give two names of molecular modeling software. | 2 |
| G) Define COMSIA with its two applications | 2 |
| H) Define Pharmacophore. | 2 |
| I) What is CADD? | 2 |
| J) Give applications of Free Wilson analysis. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- | | |
|---|----|
| A) Describe Craig Plot with the diagram. | 10 |
| B) Define QSAR and explain the requirements for a QSAR study with detail references to descriptors. | 10 |
| C) Explain molecular modeling in the context of historical times and in the applications in modern times. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- | | |
|--|---|
| A) Enlist the various databases applications in drug design? | 5 |
| B) What properties a lead compound should possess to develop as an orally active compound | 5 |
| C) Discuss in detail on various parameters used for the validation of Pharmacophore model | 5 |
| D) Enumerate the steps involved in homology modeling with a brief discussion on each step. | 5 |
| E) Write a note on Alphafold. How it is different from traditional homology modeling? | 5 |
| F) Explain Bioisosterism. Classify with examples. | 5 |
| G) Discuss in detail the various statistical tools in building PK predictive models? | 5 |
| H) Briefly discuss of defects in PDB files of proteins. | 5 |
| I) Discuss the Pharmacophore modeling protocol in detail. | 5 |

*** * * End * * ***